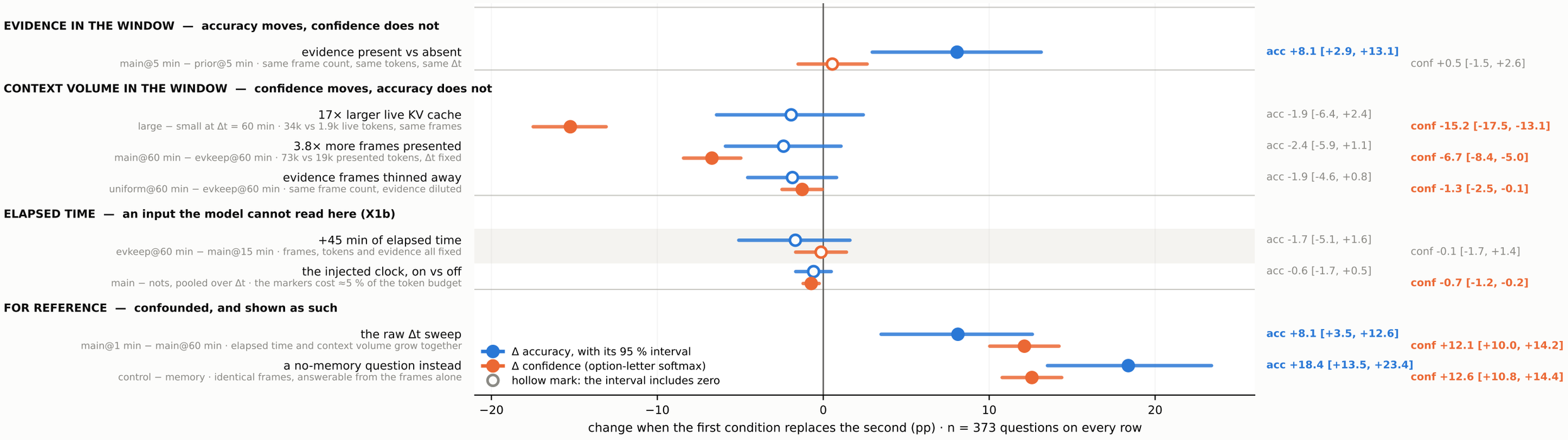
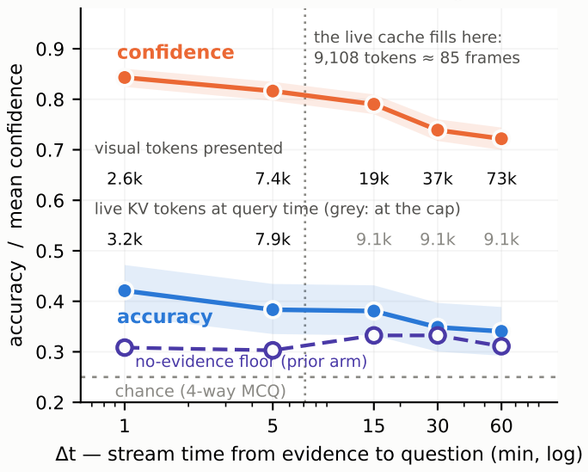
Accuracy tracks the evidence; confidence tracks how full the cache is

Streaming KV cache (EgoStream FlexibleCache + ReKV retrieval) over a frozen Qwen3-VL-8B-Instruct · 373 EgoLifeQA questions, 0.2 fps frames at 320×320 (100 visual tokens/frame) · 17,311 scored records
Confidence is max softmax over the option-letter logits {A,B,C,D} at the first generated position — a model output, not a calibrated probability. Every Δ is a paired within-question contrast, cluster-bootstrapped over questions (5,000 draws, seed 0); bands and bars are 95 % intervals. The cache is reset per question, so this is streaming over the window, not over the life-log.

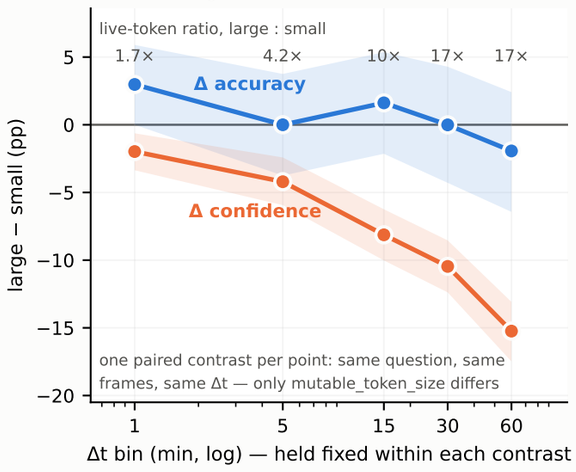
A What each manipulation moves. Accuracy responds to whether the evidence is in the window and to nothing else; confidence responds to how much context is in the window and to nothing else — the two measures are driven by disjoint variables



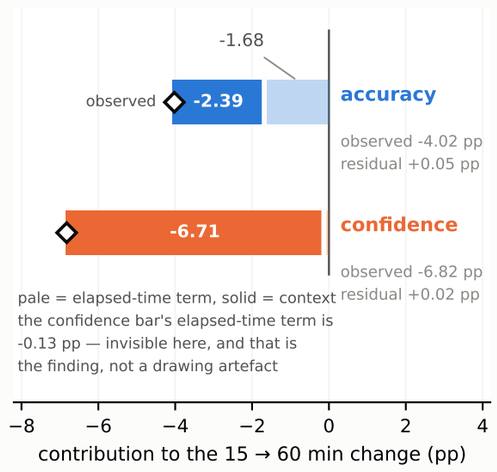
B The sweep alone settles nothing: both candidate causes rise with Δt together



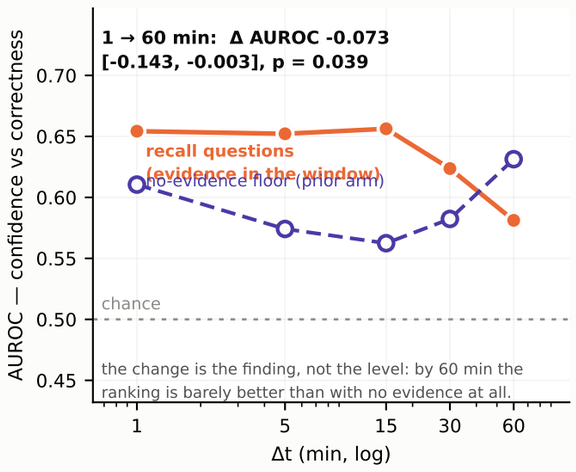
C A bigger cache buys no accuracy and costs a great deal of confidence



D The same split arithmetically: the decline is all context, with no time term left over



E With real eviction the confidence ordering decays too, not only its level



Arms · main: the Δt window ending at the question, evidence inside it prior: an evidence-free window of the same frame count (the floor) evkeep / uniform: the Δt = 15 min frame count re-drawn at Δt = 60 min, evidence frames kept vs uniformly thinned
control: a no-memory question answerable from the same frames small / large: mutable_token_size 1,024 / 32,768 against the 8,192 default (live caps 1,940 / 33,684 vs 9,108) nots: the injected clock removed
Read with care · the elapsed-time rows of A are an unavailable axis, not a null: X1b established, before this sweep ran, that the injected clock is illegible once the cache saturates — so this experiment cannot test whether confidence tracks elapsed time, only that this cache leaves it nothing to track.
In C, the large and small arms are both at their caps by Δt = 30 min, so the live-token ratio stops growing and what still grows is eviction pressure. Levels are not comparable with the full-reprocess sweep (different system prompt, different ingestion path); contrasts are.
The no-evidence floor is measured at every Δt on the grid (1, 5, 15, 30, 60 min); nothing about it is assumed. One participant (EgoLife A1), one model, one cache policy (EgoStream's shipped pruning pipeline plus ReKV retrieval).
Not shown, not a finding · item-level metamemory: among items correct at the earlier bin, confidence fell -3.0 pp on the 465 kept and -6.8 pp on the 107 lost; the difference is -3.8 [-8.5, +1.1] pp, p = 0.118 — suggestive, underpowered.
Sources · artifacts/x9_regime_comparison.json (contrasts, budget arm, per-cell levels) and artifacts/m7_metrics_stream.json (per-bin levels, decomposition, AUROC, item transitions). Drawn by egovm/streaming/x10_figure.py; no number is computed here.